Lab 39 — Phishing Recognition and Social Engineering Defence

> **Course:** CompTIA Certified A+ Training (Core 1 and Core 2) (TGS-2024048317)
> **Topic 07:** Security — Core 2, 28% of Core 2
> **Exam objective:** Identify phishing indicators and social engineering tactics, and design the user awareness training that defends against them (Core 2 objectives 2.4 and 2.7).

Goal

You work the Cybersecurity Simulator's phishing and social engineering modules to build recognition from real examples, then convert what you learned into the awareness training an organisation would actually deliver.

What you'll produce

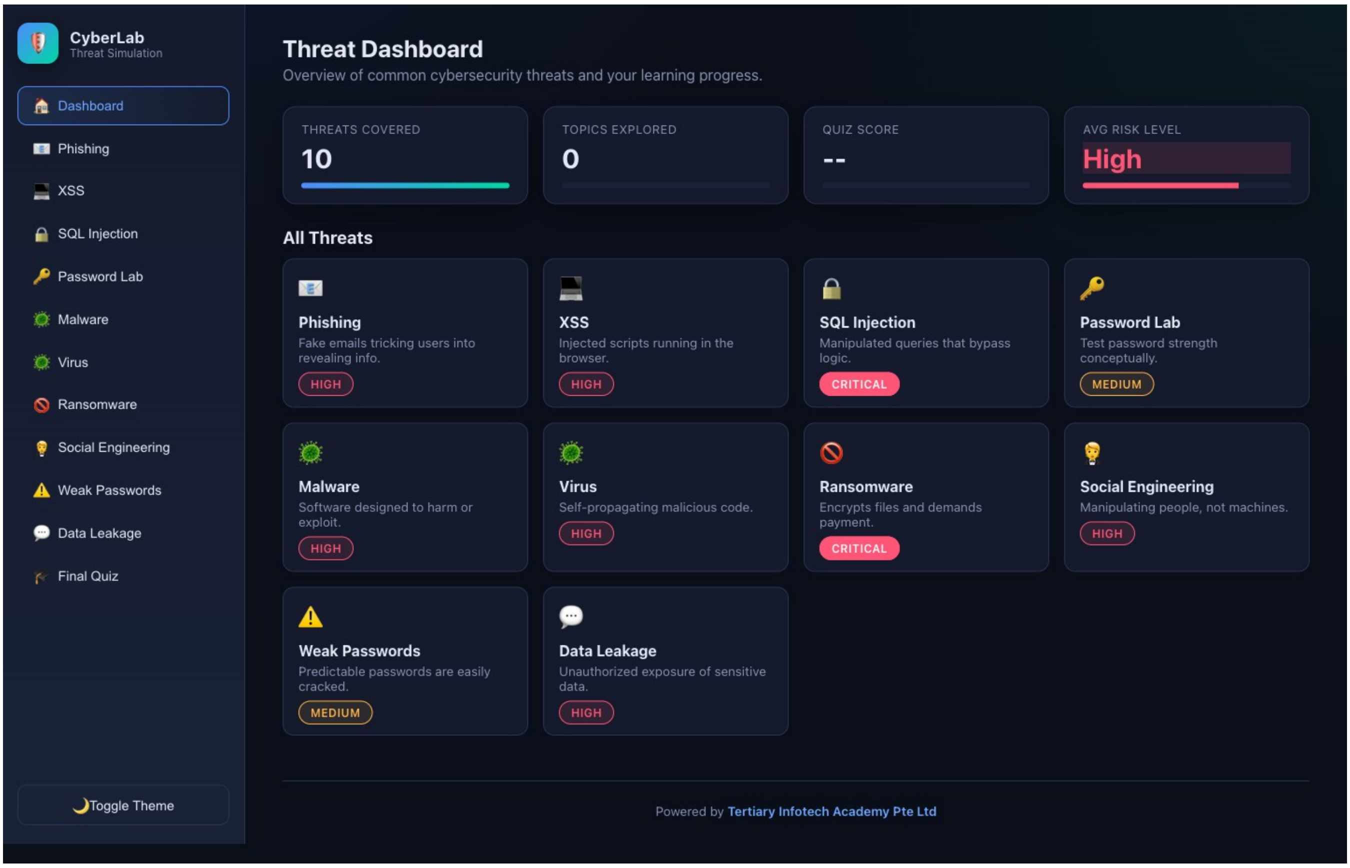
A phishing indicator checklist, a social engineering tactic reference and a user awareness training outline.

Tools and equipment

Cybersecurity Simulator (https://alfredang.github.io/cybersecuritysimulator/) — Phishing and Social Engineering modules

Browser tools used in this lab

- **Cybersecurity Simulator** — https://alfredang.github.io/cybersecuritysimulator/



Cybersecurity Simulator — the panels and fields this lab uses.

Safety

- Disconnect mains power and hold the power button for 15 seconds before working inside any machine.
- Wear an anti-static strap connected to an earthed point; hold boards and cards by their edges.
- **Never open a power supply unit or a CRT monitor** — both retain a lethal charge after being unplugged.
- Never puncture, compress or charge a swollen lithium battery; isolate it and follow hazardous disposal procedure.

Step-by-step

1. Open https://alfredang.github.io/cybersecuritysimulator/ and select Phishing from the top menu.
2. Classify at least ten emails as safe or phishing using the category filter to cover Banking, Tech, Workplace, Delivery, Social Media and Lottery.
3. Record your score and, for every email you classified wrongly, record the indicator you missed.
4. Use the annotated walkthrough feature to review the red flags on a phishing example, and record every flag it highlights.
5. Paste a suspicious link into the URL inspector and record what the analysis reveals about the actual destination against the displayed text.
6. Build the phishing indicator checklist: mismatched sender domain, generic greeting,

manufactured urgency, unexpected attachment, link text differing from destination, spelling and grammar errors, and a request for credentials.
7. Complete the phishing quick quiz and record your score and any concept you needed to revisit.
8. Switch to the Social Engineering module and work through the scenario decisions, recording your score.
9. Use the tactic filter to study each tactic in turn: pretexting, baiting, tailgating, vishing, smishing, business email compromise, quid pro quo and watering hole.
10. Build the tactic reference with a definition, a realistic example and the specific defence for each of the eight tactics.
11. Distinguish the phishing variants precisely: phishing is broad and untargeted, spear phishing targets a named individual, whaling targets a senior executive, vishing uses voice and smishing uses SMS.
12. Write the training outline: session objectives, the scenarios you would present, how you would test retention, and the reporting procedure a user must follow when they suspect an attack.

Test it — verification

Your checklist covers at least seven phishing indicators; the tactic reference defines and defends all eight social engineering tactics; the phishing variants are correctly distinguished by target and channel; and the training outline includes a reporting procedure.

Troubleshooting this lab

Symptom	What to check
A command returns "command not found"	Re-run the `apt-get install` step at the start of the lab — the playground starts with a minimal package set.
The Killercoda terminal has reset	The playground times out when idle. Reopen it and re-run the setup commands from step 1.
A browser tool will not load	Check the URL against `labs/tools.md`. All four tools run entirely client-side and need no login.
Output differs from the guide	Record what you actually observed — your environment differs from the reference, and explaining the difference is part of the exercise.

Review questions

1. State the exam objective this lab maps to, in your own words.
2. Which single step in this lab would you perform first on a real support call, and why?
3. What evidence would you attach to a support ticket to show this work was completed correctly?
4. Name one thing that would make this procedure fail, and how you would recognise it.

Record your evidence

Complete worksheet.md as you work through this lab and keep it — the Practical Performance assessment mirrors these tasks.

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